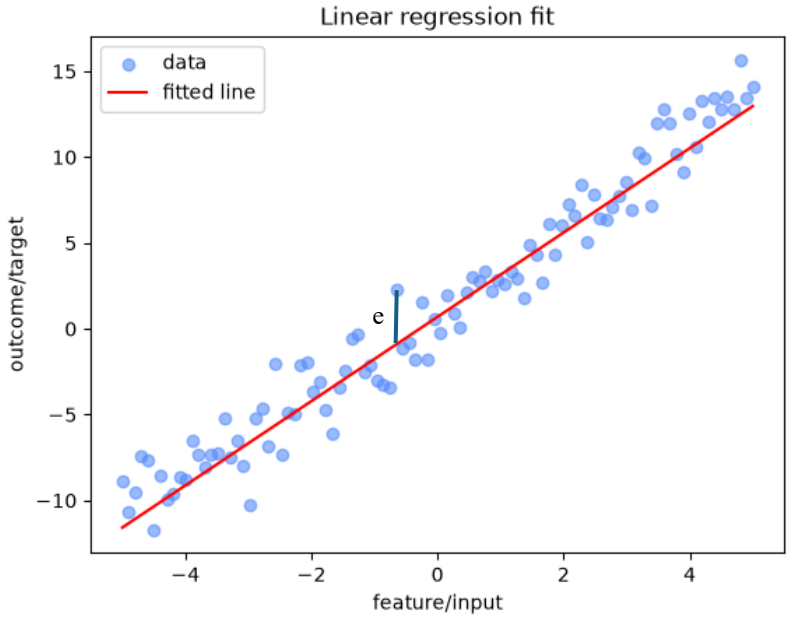
Introduction to Machine Learning

Session 3 – Lecture Notes

Linear Regression

$$y = wx + b$$



$$Error: e = y_i - (wx_i + b)$$

Loss function for linear regression:

- MSE — Mean Square Error

$$L = \frac{1}{n} \cdot \sum_{i=1}^{n} e^2 = \frac{1}{n} \cdot \sum_{i=1}^{n} (y_i - (wx_i + b))^2$$

- MAE — Mean Absolute Error

$$L = \frac{1}{n} \cdot \sum_{i=1}^{n} |e| = \frac{1}{n} \cdot \sum_{i=1}^{n} |y_i - (wx_i + b)|$$

MSE finds the model that predicts the conditional mean.

MAE finds the model that predicts the conditional median.

Gradient Descent – MSE

$$e = y - \hat{y} = y - wx - b$$

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\frac{\partial L}{\partial e} = \frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial e}\frac{\partial L}{\partial w} = \frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)(-x_i)$$

$$= -\frac{2}{n}\sum_{i=1}^{n}x_i(y_i - \hat{y}_i)$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial e}\frac{\partial L}{\partial b} = \frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)(-1)$$

$$= -\frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)$$

$$\nabla L = \left(-\frac{2}{n}\sum_{i=1}^{n}x_i(y_i - \hat{y}_i)\ , -\frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)\right)$$

Closed solution:

$$\hat{y} = Xw$$

where X is homogenous coordinate, and w includes bias

$$w = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}y$$

Gradient Descent – MAE

$$e = y - \hat{y} = y - wx - b$$

$$L = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad where: \frac{\partial |x|}{\partial x} = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$ sign function

$$\frac{\partial L}{\partial e} = \frac{1}{n}\cdot\sum_{i=1}^{n}sign(y_i - \hat{y}_i)$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial e}\frac{\partial L}{\partial w} = -\frac{1}{n}\sum_{i=1}^{n}x_i\ sign(y_i - \hat{y}_i)$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial e}\frac{\partial L}{\partial b} = -\frac{1}{n}\sum_{i=1}^{n}sign(y_i - \hat{y}_i)$$

$$\nabla L = \left(-\frac{1}{n}\sum_{i=1}^{n}x_i\ sign(y_i - \hat{y}_i)\ , -\frac{1}{n}\sum_{i=1}^{n}sign(y_i - \hat{y}_i)\right)$$

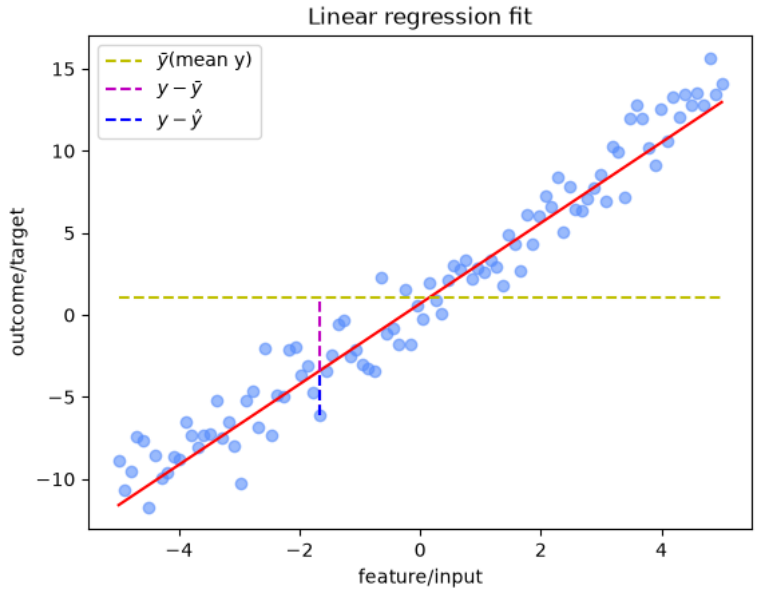
Update

$w_{new} = w_{old} - \mu\ \frac{\partial L}{\partial w}$, so it could be minimized to $w -= \mu\frac{\partial L}{\partial w}$

$b_{new} = b_{old} - \mu\ \frac{\partial L}{\partial b}$, so it could be minimized to $b -= \mu\frac{\partial L}{\partial b}$

Evaluating Regression ('Scoring Regression')

$$R^2 = 1 - \frac{SSres}{SStot} = 1 - \frac{\sum_{i=1}^{n}((y_i - \hat{y}_i)^2)}{\sum_{i=1}^{n}((y_i - \bar{y}_i)^2)}$$



$R^2 = 1$ perfect prediction (impossible)

$R^2 = 0$ there is no value in the model

$R^2 < 0$ wrong model, wrong model family

Fit percentage = 100 × R^2

Polynomial 'Linear' Regression – Feature Engineering

We expand the family to $y = w_1x^2 + w_2x + b$.

Input features are changed into [x^2, x, 1] so that the model remains linear for w_1, w_2, b.

MSE

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - w_1{x_i}^2 - w_2{x_i}^2 - b)^2$$

$$\frac{\partial L}{\partial w_1} = -\frac{2}{n}\sum_{i=1}^{n}{x_i}^2 e_i$$

$$\frac{\partial L}{\partial w_2} = -\frac{2}{n}\sum_{i=1}^{n}x_i e_i$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n}\sum_{i=1}^{n}e_i$$

$$\nabla L = \left(-\frac{2}{n}\sum x^2 e, -\frac{2}{n}\sum xe, -\frac{2}{n}\sum e\right)$$

MAE

$$L = \frac{1}{n}\sum_{i=1}^{n}|y_i - w_1{x_i}^2 - w_2{x_i}^2 - b|$$

$$\frac{\partial L}{\partial w_1} = -\frac{2}{n}\sum_{i=1}^{n}{x_i}^2 sign(e_i)$$

$$\frac{\partial L}{\partial w_2} = -\frac{2}{n}\sum_{i=1}^{n}x_i sign(e_i)$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n}\sum_{i=1}^{n}sign(e_i)$$

$$\nabla L = \left(-\frac{2}{n}\sum x^2 sign(e), -\frac{2}{n}\sum xsign(e), -\frac{2}{n}\sum sign(e)\right)$$

In principle, any function can be inserted here.